Generative AI for Work & Research

COURSE DESCRIPTION

Generative AI for Work & Research Productivity is an 8-week online course delivered by The GRAPH Courses, based at the Infectious Diseases and Mathematical Modeling group at the University of Geneva, Switzerland.

The course teaches professionals and researchers how to use generative AI tools intelligently for workflow automation, data analysis, content creation, and research synthesis without requiring programming expertise. Students apply their knowledge through hands-on workshops and a comprehensive final project demonstrating real-world AI implementation in their field of work or research.

Further details on the course can be found here: thegraphcourses.org/aiw.

WORKLOAD AND RECOMMENDED CREDITS

The 8-week course comprises weekly workshops, independent learning, and a capstone project, totaling **approximately 76 hours** of work, **corresponding to 3 ECTS**. Universities may collaborate with GRAPH to offer these credits.

A detailed breakdown of the course workload is shown below.

COURSE COMPONENT	Total Hours Dedicated Over 8 Weeks
Live workshop sessions (2 hours per week)	16
Post-workshop completion tasks (2 hours per week)	16
Prework content study (2.5 hours per week)	20
Independent work on the capstone project	24
TOTAL TYPICAL WORKING TIME: 76 HOURS	

O. Keiser

PROFESSOR OLIVIA KEISER
UNIVERSITY OF GENEVA
GRAPH NETWORK DIRECTOR

Course Syllabus

Generative AI for Work and Research: No-Code AI Tools for Task Automation

Weekly workshop: Sundays, 5–7 PM GMT **OR** Wednesdays, 3–5 PM GMT
Help sessions (optional attendance): Thursdays, 5–7 PM GMT

Learning Components

1. Prework

You will be assigned ~2 hours of prework tasks per week, to be completed before the workshop.

- 📹 **Lessons:** Work along with the video tutorials
- 📝 **Practice quizzes:** Theory questions

2. Workshops

Class time is primarily a workshop where you engage in active exercises that apply the prework while interacting with other students and instructors. Workshops last 2 hours but may require an additional 1-2 hours of work outside of class to be completed.

- 👩‍🏫 **Whole-class teaching:** Instructors demo new concepts
- 🛠️ **Learning activities:** Build skills through active learning activities

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Course Schedule

Week	Prework Topics	Workshop Topics
1. Websites & Presentations	• How LLMs are trained • How to use ChatGPT • LLM prompting strategies	• Websites with ChatGPT and GitHub Pages • Websites vs web applications • Intro to AI presentation tools
2. Automated Data Analysis	• LLM data privacy • Setting up R and Cursor	• Julius and Cursor for no-code data analyses • Bricks AI for dashboards • Shortcut AI for spreadsheets

Week	Prework Topics	Workshop Topics
3. Text Processing & Research	• Setting up to use LLMs in spreadsheets	• Text classification, sentiment analysis, and summarization with LLMs • AI tools for literature search, text analysis, and knowledge synthesis
4. AI for Workflow Automation	• AI agents vs workflows	• Building AI workflow automations • Intro to browser control agents
5. Custom GPTs and Chatbots	• LLM model terminology • Picking the right model for your task	• Creating interactive chatbots with Botpress • Building custom GPTs for specific use cases
6. GenAI for Communication and Outreach	• How models understand & create images and videos	• Graphics with ChatGPT and Canva • Editable diagrams with AI-generated SVG • Intro to AI-generated videos
7. Final Project Feedback	• Prep for final projects	• Progress check on projects • Peer feedback sessions • Troubleshooting technical challenges
8. Final Project Showcase	• Work on final projects	• Final project presentations • Public Q&A session

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Grading Rubric

The final grade will be a combination of the following:

- Graded assignments from the workshop (40%)
- Workshop attendance and participation (10%) (Attend at least 5 workshops to receive full attendance credit)
- Final project (50%)

Passing grade: 80%

Generative AI for Work & Research

No-Code AI Tools for Task Automation

An 8-week Live Cohort Course

TABLE OF CONTENTS

BOOTCAMP FEATURES

GENERATIVE AI FOR WORK & RESEARCH PRODUCTIVITY

NO-CODE AI TOOLS FOR TASK AUTOMATION

A 2-**MONTH** GUIDED COURSE

- Weekly live workshops
- Live help sessions every week
- Private community forums
- Personalized feedback on assessments
- Capstone project coaching
- Shareable certificate
- Lifetime access to our tutorial library

MODULE 1

WEBSITES, DOCUMENTS & PRESENTATIONS

Learn how large language models (LLMs) work. Generate text for papers or grants, create project websites with no-code tools, format documents efficiently, and design clear presentations or posters using AI assistants.

Topics:

- How LLMs are trained and an overview of popular LLM providers
- Generating Markdown, LateX and HTML outputs
- Building simple research websites with ChatGPT Canvas and Replit
- Using Markdown for structured research documents
- Introduction to AI presentation tools for scientific communication

MODULE 2

AI-AUTOMATED DATA ANALYSIS

Master AI tools for analyzing data, manipulating files, and creating automated reports. Use Julius, ChatGPT Code Interpreter and Windsurf to process datasets, visualize results, and generate reports quickly.

Topics:

- Use ChatGPT Code Interpreter for quick data analysis
- Build automated dashboards with TheBricks
- Process and transform datasets with Julius
- Implement batch file conversion and manipulation
- Create automated reports and data visualizations
- Design spreadsheet automations for recurring analyses

MODULE 3

GENAI FOR RESEARCH & TEXT ANALYSIS

Learn AI techniques for literature search, text analysis, and knowledge synthesis. Explore how AI can accelerate information gathering, identify patterns in text data, and organize complex information. Develop systems for extracting insights from large text collections and academic publications.

Topics:

- Conducting efficient literature reviews with Elicit and LiteRev
- Implementing text classification for document organization
- Applying sentiment analysis to identify trends and opinions
- Creating knowledge maps from document collections
- Designing automated summarization workflows
- Building information synthesis systems for research topics

MODULE 4

CUSTOM GPTS AND CHATBOTS

Dive deeper into research assistant creation by leveraging advanced LLM architecture. Build chatbots for research assistance and knowledge management. You'll gain insights into context windows, retrieval augmentation, and how to integrate AI assistants into your research workflow.

Topics:

- LLM chat architecture, context windows, and RAG systems
- Building custom GPTs for specific use cases
- Creating interactive chatbots with Botpress
- Integrating knowledge bases with your research materials

MODULE 5

AI FOR TASK AUTOMATION

Discover practical strategies for automating repetitive research tasks through AI-driven flows. This module empowers you to set up email response automations, design AI agents for literature monitoring, and orchestrate complex workflows that save time and resources.

Topics:

- Introduction to AI automation and agent-based systems
- Setting up email response automations
- Building AI agents for complex automation flows
- Browser control agents for web-based tasks

MODULE 6

GENAI FOR COMMUNICATION AND OUTREACH

Explore AI-driven approaches for research dissemination, focusing on creating assets for sharing your findings. Learn to develop methods for summarizing research, design visuals for publications or presentations, and produce explanatory content with minimal manual effort.

Topics:

- AI-generated research summaries for different audiences
- Designing scientific figures and diagrams with AI tools
- AI-generated video abstracts and explanations
- Prompt engineering for scientific visualization
- Content scheduling for research communication

MODULE 7

CAPSTONE PROJECT & SHOWCASE

Collaborate with peers and instructors to create and refine a final AI solution, tool or presentation. Receive real-time feedback and troubleshoot technical challenges. Conclude the program by presenting your final projects to an audience of peers and guests.

Topics:

- Project progress check
- Peer feedback sessions
- Final project presentations
- Public Q&A session

GRADING RUBRIC

The final course grade will be a combination of the following:

- Online quizzes (15%)
- Workshop assignments (35%)
- Workshop attendance & participation (10%)
- Final project (40%)

Passing grade: 80%

Attendance policy: Attend at least 5 workshops to receive full attendance credit

WHY GRAPH COURSES?

At GRAPH Courses, we're committed to making high-quality data science and AI education accessible and impactful. Based at the **University of Geneva, Switzerland**, our programs combine academic rigor with practical, industry-relevant skills.

Our bootcamp-style approach ensures an immersive learning experience. You'll gain technical know-how, build **real-world AI projects**, and set yourself apart in the job market.

What sets us apart is our supportive learning ecosystem. With live workshops, personalized project reviews, and a global community of peers, you're never alone. Our experienced instructors and mentors are dedicated to your success.

As a non-profit backed by academic institutions, we offer this comprehensive bootcamp at an affordable price without compromising on quality.

INSTRUCTORS & MENTORS

Prof. Olivia Keiser
GRAPH Network Director
Head of the Infectious Diseases & Mathematical Modelling Division, University of Geneva

Kene Nwosu
GRAPH Instructor
Data Analyst & PhD student, Generative AI researcher, University of Geneva

Prof. Flavio Coelho
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GRAPH

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Geneva Health Forum

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